

Reviewer Pack — Minimal Rank of Quandle Representations Bounds Resolution Pr...

Entry 79a2459f4b63 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Quandle Representations Bounds Resolution Proof Width

Entry ID: 79a2459f4b63 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-29 12:05:12 UTC

1. Conjecture

Field A (mathematical branch): Quandle Theory **Field B** (complexity object): Resolution proof complexity

Statement:

For a given Boolean function f represented by a quandle Q , the width of the resolution proof for f is bounded by $O(\log^2(n) * \log(r(Q)))$, where $r(Q)$ is the rank of the quandle Q .

Rationale (proposer's reasoning):

Quandle theory provides a categorical representation of Boolean functions that can be used to capture the structure of proofs in propositional logic. A lower bound on the resolution proof width would imply a relationship between the algebraic properties of quandles and the complexity of resolution proofs, which could lead to new insights into the nature of computational hardness.

Taxonomy category: c003b_cumulative_entropy (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 00b59580a5993700

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if at least 80% of quandle representations Q (with rank $r(Q)$) for Boolean functions f have resolution proof widths $\leq O(\log^2(n) * \log(r(Q)))$. The conjecture is falsified if any seed produces a quandle representation with resolution proof width $> O(\log^2(n) * \log(r(Q)))$ or if less than 80% of the quandle representations meet the support condition.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	SAFE
NATURAL_PROOFS	SAFE	0.70	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=0, elapsed=0.5s

5.1 Generated Python source

```
#!/usr/bin/env python3
"""C-003b cumulative entropy – FAITHFUL reference harness.

Computes the REAL cumulative proof entropy
    Phi(pi) = sum_t |activeClauses(sigma_t)|
over an actual resolution refutation (Davis-Putnam variable elimination)
of Tseitin formulas, exactly per the Lean definitions in Conjecture003.lean:
    proofStateEntropy(sigma) := sigma.activeClauses.card
    cumulativeEntropy(states) := sum (map proofStateEntropy states)
plus the width-aware totalLiteralWeight version (sum of clause sizes).

This is NOT the CDCL clause-count proxy used by the old c003b_counterexample.py:
it tracks the active clause database at EVERY elimination step of a genuine
resolution derivation, so it is the actual Phi the formalization names.

Pure stdlib (no networkx/pysat), so it runs in the pipeline sandbox.
Protocol: run_trial(seed)->dict with TRIAL/RESULT lines.

Author: Ludovico Kubler (harness by the SEC engine, hand-verified).
NOTE: no `from __future__ import annotations` – the sandbox injects a prelude
above the file, which would break that statement's must-be-first rule.
"""

import json
import math
import random
import sys

Clause = frozenset # a clause = frozenset of signed ints (var or -var)

# ----- Tseitin formula generation -----

def random_regular_graph(n: int, degree: int, rng: random.Random):
    """Simple random regular graph via configuration model with retries."""
    if (n * degree) % 2 != 0:
        n += 1
    for _ in range(200):
```

```

stubs = []
for v in range(n):
    stubs += [v] * degree
rng.shuffle(stubs)
edges = set()
ok = True
for i in range(0, len(stubs), 2):
    u, w = stubs[i], stubs[i + 1]
    if u == w or (min(u, w), max(u, w)) in edges:
        ok = False
        break
    edges.add((min(u, w), max(u, w)))
if ok and len(edges) == n * degree // 2:
    return n, sorted(edges)
# fallback: cycle + chords
edges = {(i, (i + 1) % n) for i in range(n)}
return n, sorted((min(a, b), max(a, b)) for a, b in edges)

def tseitin_cnf(n: int, edges: list, charge: dict) -> list:
    """Tseitin CNF over edge variables (1-indexed). For each vertex, clauses
    enforce parity of incident edges = charge[v]."""
    evar = {}
    for i, (u, v) in enumerate(edges):
        evar[(u, v)] = i + 1
        evar[(v, u)] = i + 1
    inc = {v: [] for v in range(n)}
    for (u, v) in edges:
        inc[u].append(evar[(u, v)])
        inc[v].append(evar[(u, v)])
    clauses = []
    for v in range(n):
        vars_ = inc[v]
        k = len(vars_)
        tgt = charge.get(v, 0)
        for mask in range(1 << k):
            if bin(mask).count("1") % 2 != tgt:
                cl = []
                for j, var in enumerate(vars_):
                    cl.append(-var if (mask >> j) & 1 else var)
                clauses.append(frozenset(cl))
    return clauses

# ----- Resolution by Davis-Putnam variable elimination -----

def resolve(c1: frozenset, c2: frozenset, var: int):
    """Resolvent of c1,c2 on var (c1 has +var, c2 has -var). None if tautology."""
    r = (c1 - {var}) | (c2 - {-var})
    for lit in r:
        if -lit in r:
            return None # tautology
    return frozenset(r)

def dp_refutation_phi(clauses: list, rng: random.Random):

```

"""Run DP elimination in min-occurrence order; track the active clause DB after each elimination. Returns (phi_count, phi_weight, n_steps, derived_empty)."""

```
phi_count = sum_t |DB_t| (proofStateEntropy = card)
phi_weight = sum_t sum_{C in DB_t} |C| (totalLiteralWeight version)
"""
```

```
db = set(clauses)
variables = set(abs(l) for c in db for l in c)
phi_count = len(db)
phi_weight = sum(len(c) for c in db)
n_steps = 1
derived_empty = frozenset() in db
MAX_DB = 200000 # guard against blow-up on bad instances
```

```
while variables and not derived_empty:
    # min-occurrence elimination order (standard good heuristic)
    occ = {v: 0 for v in variables}
    for c in db:
        for l in c:
            if abs(l) in occ:
                occ[abs(l)] += 1
    var = min(variables, key=lambda v: occ[v])
    variables.discard(var)

    pos = [c for c in db if var in c]
    neg = [c for c in db if -var in c]
    others = [c for c in db if var not in c and -var not in c]

    resolvents = set()
    for cp in pos:
        for cn in neg:
            r = resolve(cp, cn, var)
            if r is not None:
                resolvents.add(r)
                if len(r) == 0:
                    derived_empty = True
    new_db = set(others) | resolvents
    # forward subsumption (keep minimal clauses) – cheap pass
    db = new_db
    phi_count += len(db)
    phi_weight += sum(len(c) for c in db)
    n_steps += 1
    if len(db) > MAX_DB:
        break

return phi_count, phi_weight, n_steps, derived_empty
```

----- Separator (cheap balanced-ish) -----

```
def approx_separator_size(n: int, edges: list) -> int:
    """Cheap proxy for balanced separator size: min vertices whose removal
    splits the graph ~in half. We use a simple BFS-layer cut."""
    adj = {v: set() for v in range(n)}
    for (u, v) in edges:
        adj[u].add(v); adj[v].add(u)
```

```

# BFS from 0, cut at the layer crossing the median
from collections import deque
seen = {0}; layer = {0: 0}; q = deque([0])
while q:
    x = q.popleft()
    for y in adj[x]:
        if y not in seen:
            seen.add(y); layer[y] = layer[x] + 1; q.append(y)
if len(seen) < n: # disconnected
    return 0
half = n // 2
# vertices on the boundary between the first ~half and the rest
order = sorted(range(n), key=lambda v: layer.get(v, 0))
side_a = set(order[:half])
cut = set()
for (u, v) in edges:
    if (u in side_a) != (v in side_a):
        cut.add(u if u not in side_a else v)
return len(cut)

```

----- Trial protocol -----

```

def run_trial(seed: int) -> dict:
    rng = random.Random(seed)
    # sweep several small sizes; report the scaling exponent of Phi vs n
    ns = [6, 8, 10, 12, 14]
    data = []
    for n0 in ns:
        n, edges = random_regular_graph(n0, 3, rng)
        charge = {v: 0 for v in range(n)}
        charge[0] = 1 # odd total -> UNSAT
        clauses = tseitin_cnf(n, edges, charge)
        phi_c, phi_w, steps, empty = dp_refutation_phi(clauses, rng)
        sep = approx_separator_size(n, edges)
        data.append({"n": n, "m_edges": len(edges), "sep": sep,
                    "phi_count": phi_c, "phi_weight": phi_w,
                    "steps": steps, "derived_empty": empty})
    # scaling: log-log slope of phi_count vs n
    xs = [math.log(d["n"]) for d in data if d["phi_count"] > 0]
    ys = [math.log(d["phi_count"]) for d in data if d["phi_count"] > 0]
    if len(xs) >= 2:
        mx = sum(xs) / len(xs); my = sum(ys) / len(ys)
        num = sum((x - mx) * (y - my) for x, y in zip(xs, ys))
        den = sum((x - mx) ** 2 for x in xs)
        slope = num / den if den else 0.0
    else:
        slope = 0.0
    biggest = data[-1]
    # Sub-conjecture under test (SC1): Phi >= sep^2 on the largest instance
    # (a concrete, falsifiable scaling claim derived from the C-003b thesis
    # that cumulative entropy is forced by the separator).
    holds = biggest["phi_count"] >= max(1, biggest["sep"]) ** 2
    return {
        "metric_name": "phi_count_loglog_slope_vs_n",
        "metric_value": round(slope, 4),
    }

```

```

    "instances_tested": len(ns),
    "n_max": max(ns),
    "conjecture_holds": holds,
    "counterexample": "" if holds else
        f"n={biggest['n']} sep={biggest['sep']} phi={biggest['phi_count']} < sep^2",
    "detail": data,
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [11, 23, 37]
    results = []
    for s in seeds:
        r = run_trial(s)
        results.append(r)
        print("TRIAL: " + json.dumps(r))
    slopes = [r["metric_value"] for r in results]
    holds = sum(1 for r in results if r["conjecture_holds"])
    mean = sum(slopes) / len(slopes)
    if holds == len(results):
        print(f"RESULT: SUPPORTED mean_slope={mean:.4f} support_fraction=1.0")
    elif holds == 0:
        print(f"RESULT: FALSIFIED counterexample=\"phi < sep^2\" first_failing_seed={seeds[0]}")
    else:
        print(f"RESULT: INCONCLUSIVE mixed holds={holds}/{len(results)} mean_slope={mean:.4f}")

```

6. Per-seed results

Seed	Metric value	Holds?	Counterexample
?	2.4348	□	
?	2.0177	□	
?	2.0528	□	
?	2.4751	□	
?	2.4691	□	
?	2.2173	□	
?	2.2983	□	
?	2.7629	□	
?	2.3782	□	
?	2.7571	□	
?	2.6234	□	
?	2.5729	□	
?	2.3297	□	

Aggregate statistics:

Statistic	Value
n_seeds	13
metric_mean	2.4145615384615384
metric_std	0.23518233897149377
metric_ci95_half	0.13045568957619594
metric_min	2.0177
metric_max	2.7629
support_fraction	1.0

7. Test stdout (last 2KB)

```
"m_edges": 15, "sep": 3, "phi_count": 591, "phi_weight": 2826, "steps": 16, "derived_empty": true}, {"  
TRIAL: {"metric_name": "phi_count_loglog_slope_vs_n", "metric_value": 2.2983, "instances_tested": 5,  
TRIAL: {"metric_name": "phi_count_loglog_slope_vs_n", "metric_value": 2.7629, "instances_tested": 5,  
TRIAL: {"metric_name": "phi_count_loglog_slope_vs_n", "metric_value": 2.3782, "instances_tested": 5,
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

The test code provided does not implement the conjecture’s statement regarding quandle representations and resolution proof width. The code focuses on computing cumulative proof entropy for Tseitin formulas using Davis-Putnam variable elimination, which is unrelated to quandles or their rank. There is no evidence of a quandle representation or its rank being used in the test.

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The critic challenged the validity of the test code as it does not implement the conjecture’s statement regarding quandle representations and resolution proof width. | next: Implement a test that correctly uses quandle representations and their ranks to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 6

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	23316
2	preregistration	ollama_remote	glm4:latest	0	0	15726
3	novelty	ollama_remote	glm4:latest	0	0	9140
4	novelty	ollama_remote	glm4:latest	0	0	9615
5	critic	ollama_remote	glm4:latest	0	0	17675
6	judge	ollama_remote	glm4:latest	0	0	10146

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 85617 ms total latency. Provider mix: {'ollama_remote': 6}

(full prompt+response transcripts available in `research/audit/79a2459f4b63.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/79a2459f4b63.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/79a2459f4b63.tar.gz (if generated)